

JNTU Hyderabad

MID-TERM EXAMINATION ? 2025

Data Structures ? Mid-Term Examination, 2025

Subject Code: CS301 Semester: 3 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Arrays and Linked Lists)

1. Explain the array representation of lists. Compare array and linked list implementations with respect to memory usage and performance. [12 Marks]
2. Write algorithms to insert and delete a node at the beginning and at the end of a singly linked list. Explain with neat diagrams. [12 Marks]
3. Differentiate between singly linked list, doubly linked list and circular linked list with suitable examples. [12 Marks]

SECTION B (Unit 2: Stacks and Queues)

4. Explain the stack operations (push, pop, peek) with algorithms. What are the important applications of stacks? [12 Marks]
5. Implement a queue using a linked list. Compare linear queue, circular queue and priority queue. [12 Marks]
6. Convert the infix expression $(A+B)*C-D$ to postfix notation and evaluate it using a stack. Show all the steps. [12 Marks]

SECTION C (Unit 3: Trees and Binary Search Trees)

7. Construct a Binary Search Tree for the given set of elements and perform inorder, preorder and postorder traversals. [12 Marks]
8. Explain AVL tree rotations (LL, RR, LR, RL) with an example. Why are AVL trees preferred over simple BSTs? [12 Marks]
9. Write algorithms for insertion and deletion in a Binary Search Tree with examples. [12 Marks]

SECTION D (Unit 4: Graphs and Traversals)

10. Explain BFS and DFS graph traversal algorithms with examples and time complexity analysis. [12 Marks]
11. Explain Prim's and Kruskal's algorithms for finding the minimum spanning tree with an example. [12 Marks]
12. Solve a problem related to Graphs and Traversals. Show all the steps clearly. [8 Marks]

SECTION E (Unit 5: Hashing and Searching)

13. What is hashing? Explain collision resolution techniques ? separate chaining and open addressing (linear, quadratic, double hashing). [12 Marks]

14. Explain linear search and binary search with algorithms. Compare their time complexities with justification. [8 Marks]

15. How is Hashing and Searching implemented? Give an algorithm / procedure and explain. [8 Marks]

*** End of Question Paper ***